

THE SEPARATION CRITERION AT EVERY DIMENSION ON EVERY CUBICAL SITE WITH REVERSAL

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2020 *Mathematics Subject Classification.* 55U35; 18N40, 20F55, 06D30, 03B38.

Key words and phrases. cubical sets, cubical type theory, model structure, test category, reflection group, De Morgan algebra, Kleene algebra.

ABSTRACT.

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1. INTRODUCTION

Let $\mathcal{V}$ be one of the three standard varieties with reversal — De Morgan, Kleene, Boolean — and its cube category; presheaves on carry the cubical-type model structure (uniform Kan fibrations) and Cisinski’s test structure, with weak equivalences $W_{\text{type}} \subseteq W_{\text{test}}$. For a subgroup $H \leq B_n$ of the hyperoctahedral group, write $\text{SEP}_{\mathcal{V}}(H)$ for the statement that an explicit map in $W_{\text{test}} \setminus W_{\text{type}}$ is attached to the isotropy quotient $\square^n/H$. The companion manuscript [1] proved, on the Boolean site,

$$\text{SEP}_{\text{BA}}(H) \iff \langle\langle \text{Refl}(H) \rangle\rangle \text{ is mixed} \quad \text{for every } n \text{ and every } H \leq B_n,$$

by a sieve-landing dichotomy: the strict homotopy component of the identity of $\square^n/N$ consists exactly of twisted-equivariant vertex maps, so either none is stabilized — and the attached witness separates — or one is, and the quotient reduces to a strictly smaller instance. The Boolean proof leaned on a special feature of that site: cells are arbitrary vertex functions, so any pair of maps with common deck data assembles into a cylinder.

This paper removes the special feature and proves:

Theorem 1.1 (Main theorem). *On each of the De Morgan, Kleene, and Boolean sites, for every n and every $H \leq B_n$:*

$$\text{SEP}_{\mathcal{V}}(H) \iff \langle\langle \text{Refl}(H) \rangle\rangle \text{ is mixed,}$$

and when $\langle\langle \text{Refl}(H) \rangle\rangle$ has a common fixed cell, $\square^n/H \simeq B(H/\langle\langle \text{Refl}(H) \rangle\rangle)$ in both structures.

The De Morgan site is the site of CCHM cubical type theory, so Theorem 1.1 settles the isotropy-quotient comparison for cubical type theory itself at every dimension. The proof has three inputs, each of independent interest.

The equivariant interpolant (Section 3). The convexity interpolant $\chi(u, v, t)$, introduced in the strict-fiber theory of the early papers of this series, satisfies the unexpected identity $\chi(\neg u, \neg v, t) = \neg \chi(u, v, t)$ in free De Morgan algebras. Consequently a componentwise χ -cylinder between two cells commutes with every signed permutation, and two strict cells sharing exact deck data are one-step homotopic *on every reversal site*. This is the exact replacement for the Boolean “any pair is a cylinder,” and it makes the entire dichotomy of [1] available verbatim with “twisted vertex map” replaced by “twisted cell.”

The projection lemma (Section 4). The Boolean variety is a quotient of the De Morgan and Kleene varieties, so free-algebra surjections $\pi: F_{\mathcal{V}}(k) \twoheadrightarrow F_{\text{BA}}(k)$ induce a site functor that preserves the group actions, deck data, cylinder ends, constants, stabilization, and strict invariance. Strict chains project; hence if the Boolean instance is rigid, so is the $\mathcal{V}$ -instance, and the trichotomy exclusions transfer.

The algebraic reduction (Section 5). When a stabilized twisted $\mathcal{V}$ -cell exists, its exact stabilization equalities force its components to satisfy the pattern relations of the fixed locus of its stabilizer — an *algebraic* factorization through the pattern embedding $\iota: \square^d \rightarrow \square^n$, with no appeal to vertex data. Both composites of the induced pair of maps between $\square^n/N$ and $\square^d/\text{res}$ are twisted cells, hence one-step homotopic to identities by the interpolant: the quotient is equivalent, in both structures at once, to a lower-dimensional isotropy quotient. The reflection-transport argument of [1] keeps the residual group mixed and reflection-generated, and the criterion recurses on n — with the pleasant simplifications that no power of the cell, no eventual image, and no invertibility argument are needed.

At $n = 4$, where [1] found exactly two reduction classes, an exact solver (parity constraints plus monotonicity, formulated as 2-SAT over the literal cube) shows that both classes admit De Morgan and Kleene witnesses — for the second class on exactly one of its three invariant planes, the same plane as its Boolean witnesses (Section 7). The transfer of the $n \leq 4$ classification is therefore concrete as well as abstract.

Relation to the companion manuscript. We use from [1]: the trichotomy for the witness machines and the stage-zero visibility lemma, whose sector and first-exit proofs are site-generic (they use only the coequalizer presentation of the algebraic replacement, connections and diagonals, the contractibility and freeness of the body, and the covering classification); the covering descent theorem; the agreement half of the criterion; and the reflection-transport lemma. Precise statements are recalled where used.

2. PRELIMINARIES

We follow the conventions of [1]. $\mathcal{V}$ -cubical sets are presheaves on $([m], [n]) = F_{\mathcal{V}}(m)^n$; the free algebras are represented as monotone $\{0, 1\}$ -functions on the literal cube L_m (De Morgan), its unmixed subposet U_m (Kleene), and all functions on $\{0, 1\}^m$ (Boolean), with negation $(\neg f)(p) = 1 - f(\rho p)$. An element of B_n is a reflection iff it fixes a vertex; H is mixed iff it fixes no vertex; $N := \langle\langle \text{Ref}(H) \rangle\rangle$.

A *cell* of $\square^n$ at level k is an n -tuple over $F_{\mathcal{V}}(k)$; a strict endomorphism datum of $\square^n/N$ is a level- n cell F with the *deck condition*: for every $h \in N$ there is $d \in N$ with $F \circ \sigma_h = \sigma_d \circ F$ (an equality of cells). F is *c-twisted* if $F \circ \sigma_h = \sigma_{c^{-1}hc} \circ F$ for all h , *twisted* if c -twisted for some $c \in N$, *stabilized* if $\sigma_s \circ F = F$ for some $s \neq 1$ (equivalently, its class lands in the stabilized sieve Σ/N), and *strictly invariant* if $F \circ \sigma_h = F$ for all h . A *one-step homotopy* between strict classes is a level- $(n+1)$ cell G with the deck condition for the N -action on the $\square^n$ -factor and with prescribed restrictions at $t = 0, 1$.

3. THE EQUIVARIANT INTERPOLANT

Lemma 3.1 (Self-duality of the convexity interpolant). *In the free De Morgan algebra on u, v, t , the element*

$$\chi(u, v, t) := (u \wedge \neg t) \vee (t \wedge v) \vee (u \wedge v)$$

satisfies

$$\chi(u, v, 0) = u, \quad \chi(u, v, 1) = v, \quad \chi(\neg u, \neg v, t) = \neg \chi(u, v, t).$$

Consequently the same identities hold in every quotient variety, in particular in free Kleene and free Boolean algebras.

Proof. The first two are the defining computations of the convexity interpolant. For the third, De Morgan's laws give

$$\neg \chi(u, v, t) = (\neg u \vee t) \wedge (\neg t \vee \neg v) \wedge (\neg u \vee \neg v).$$

Distributing the first two factors: $(\neg u \vee t) \wedge (\neg t \vee \neg v) = (\neg u \wedge \neg t) \vee (\neg u \wedge \neg v) \vee (t \wedge \neg t) \vee (t \wedge \neg v)$. Meeting with the third factor and absorbing — using $t \wedge \neg t \leq t$, $\neg t$, so $t \neg t \neg u \leq \neg t \neg u$ and $t \neg t \neg v \leq t \neg v$, and $t \neg u \neg v \leq \neg u \neg v$ — leaves

$$(\neg u \wedge \neg t) \vee (t \wedge \neg v) \vee (\neg u \wedge \neg v) = \chi(\neg u, \neg v, t).$$

Identities of free De Morgan algebras hold in all quotients; the free Kleene and Boolean algebras are quotients of the free De Morgan algebra on the same generators. (The identities are also machine-verified on the literal-cube representation; artifact.) $\square$

Proposition 3.2 (Shared deck data implies one-step homotopy). *Let φ, φ' be level- n cells with $\varphi \circ \sigma_h = \sigma_{d(h)} \circ \varphi$ and $\varphi' \circ \sigma_h = \sigma_{d(h)} \circ \varphi'$ for all $h \in N$, with the same $d(h)$. Then the componentwise interpolant $G_i := \chi(\varphi_i, \varphi'_i, t)$ is a one-step homotopy between their classes, on every reversal site. Conversely, the slices of a one-step homotopy share deck data, and a one-step homotopy from the identity class forces its other slice to be a twisted cell.*

Proof. Writing $(\sigma_d \psi)_i = \neg^i \psi_{\pi(i)}$, the substitution acts on the $\square^n$ -variables only, so

$$G_i \circ (\sigma_h \times \text{id}) = \chi(\varphi_i \sigma_h, \varphi'_i \sigma_h, t) = \chi(\neg^i \varphi_{\pi(i)}, \neg^i \varphi'_{\pi(i)}, t) = \neg^i \chi(\varphi_{\pi(i)}, \varphi'_{\pi(i)}, t) = (\sigma_{d(h)} G)_i,$$

the third equality by Lemma 3.1 when $i = 1$. The restrictions at $t = 0, 1$ are φ, φ' by the interpolation identities. For the converse, restricting the deck equality of a cylinder to the two ends gives the shared deck data; and if one slice is σ_a (a representative of the identity class), the deck element at h is forced to be aha^{-1} — automorphism cells are distinguished by their vertex maps — so the other slice satisfies a full twisted equation. $\square$

Corollary 3.3 (The dichotomy on every reversal site). *All the statements of the sieve-landing dichotomy of [1] hold verbatim on $\mathcal{V}$ with “twisted vertex map” replaced by “twisted cell”: deck freedom equals stabilization ($d_1, d_2 \in D_F(h)$ iff $\sigma_{d_1^{-1}d_2} F = F$); every twisted cell is one-step homotopic to the*

identity (translate one end by Proposition 3.2); if no twisted cell is stabilized, the strict homotopy component of the identity is exactly the set of twisted classes and contains no constant, no sieve-landing class, and no strictly invariant class.

Proof. The proofs in [1] manipulate cell equalities only — never the Boolean description of cells — except for the existence of cylinders between shared-deck pairs, which is Proposition 3.2. Constants: a constant at w is twisted only if N fixes w , excluded for mixed N . Strict invariance plus twistedness forces stabilization, excluded in the rigid case. $\square$

4. THE PROJECTION LEMMA

Lemma 4.1 (Projection). *For $\mathcal{V} \in \{\text{DM}, \text{KL}\}$ the theory quotient induces surjections $\pi: F_{\mathcal{V}}(k) \twoheadrightarrow F_{\text{BA}}(k)$ commuting with substitution, hence a site functor $\rightarrow \square_{\text{BA}}$ that is the identity on objects. On cells and cylinders, π preserves: the σ -actions; deck equalities (with the same deck elements); restrictions to cylinder ends; identity, constant, stabilized, and strictly invariant cells. Consequently every strict chain of one-step homotopies on $\mathcal{V}$ projects to a strict chain on the Boolean site connecting the projected classes.*

Proof. The signed permutations and the structural maps of the sites are given by terms in $\wedge, \vee, \neg, 0, 1$, and π is a homomorphism of algebras commuting with substitution in each variable; all listed conditions are equalities between cells built from such terms, and π of an equality is an equality. A one-step cylinder is a cell with deck and end conditions, so its image is one. $\square$

$$\begin{array}{ccc}
 \text{strict chains on } \mathcal{V} : & \text{id} \xrightarrow{G^{(1)}} F_1 \xrightarrow{G^{(2)}} \cdots \longrightarrow & \xi \\
 & & \downarrow \pi \\
 \text{strict chains on BA} : & \text{id} \xrightarrow{\pi G^{(1)}} \pi F_1 \xrightarrow{\pi G^{(2)}} \cdots \longrightarrow & \pi \xi
 \end{array}$$

FIGURE 1. The projection lemma: chains of one-step homotopies project along the theory quotient, preserving deck data, ends, constants, stabilization, and strict invariance.

Proposition 4.2 (Rigidity transfers). *If no twisted Boolean vertex map for N is stabilized, then no twisted $\mathcal{V}$ -cell for N is stabilized.*

Proof. A stabilized c -twisted $\mathcal{V}$ -cell projects, by Lemma 4.1, to a stabilized c -twisted Boolean cell, i.e. a stabilized twisted vertex map. $\square$

5. THE ALGEBRAIC REDUCTION

Theorem 5.1 (Reduction on $\mathcal{V}$). *Let $N \leq B_n$ be mixed and reflection-generated, and let F be a stabilized c -twisted $\mathcal{V}$ -cell with stabilizer $S := \{s : \sigma_s F = F\} \neq 1$. Then:*

- (1) *S is normalized by N , and $\text{Fix}(S)$ is an N -invariant locus, a subcube $\square^d$ with $d < n$ embedded by a pattern $\iota : x_i \mapsto \neg^i m_{\text{cl}(i)}$;*
- (2) *the stabilization equalities force the pattern relations on the components of F , so $F = \iota \circ \bar{F}$ for a unique level- n cell $\bar{F}$ of $\square^d$, algebraically;*
- (3) *$\bar{F}$ and ι descend to $\bar{g} : \square^n/N \rightarrow \square^d/\text{res}$ and $\bar{j} : \square^d/\text{res} \rightarrow \square^n/N$, where res is the image of N in $\text{Aut}(\square^d)$, and both composites are twisted cells, hence one-step homotopic to the respective identities: $\square^n/N \simeq \square^d/\text{res}$ simultaneously in both structures;*
- (4) *$\text{res} \leq B_d$ is mixed and generated by its reflections.*

Proof. (1) For $s \in S$ and $h \in N$, twisted equivariance gives $\sigma_{(c^{-1}hc)s(c^{-1}hc)^{-1}} F = \sigma_{c^{-1}hc} \sigma_s F \sigma_h^{-1} = F$, and $h \mapsto c^{-1}hc$ is a bijection of N ; invariance of the locus follows, and the locus-pattern analysis of [1] identifies $\text{Fix}(S)$ with a pattern-embedded subcube, of dimension $d < n$ since $S \neq 1$ acts nontrivially. (2) For each $s \in S$, $\sigma_s F = F$ reads componentwise as $\neg^i F_{\pi_s(i)} = F_i$: exactly the statement that the components of F are constant up to the pattern signs along the cycles of S , i.e. satisfy the defining relations of the image of ι . Choosing one representative component per class defines $\bar{F}$ with $F = \iota \circ \bar{F}$; uniqueness because the components of ι are distinct literals, so ι is a monomorphism on cells. (3) From $\iota \circ \bar{F} \circ \sigma_h = F \sigma_h = \sigma_{c^{-1}hc} F = \iota \circ \text{res}(c^{-1}hc) \circ \bar{F}$ and monomorphy, $\bar{F}$ intertwines along $N \twoheadrightarrow \text{res}$, and $\sigma_h \circ \iota = \iota \circ \text{res}(h)$; both descend. The composite $\bar{j} \bar{g} = [F]$ is twisted by hypothesis; the composite $\bar{g} \bar{j} = [\bar{F} \circ \iota]$ satisfies $(\bar{F} \iota) \circ \text{res}(h) = \bar{F} \sigma_h \iota = \text{res}(c^{-1}hc) (\bar{F} \iota)$, so it is $\text{res}(c)$ -twisted. Every twisted cell is one-step homotopic to the identity (Corollary 3.3), and the χ -cylinders are strict cells, hence homotopies in the type and the test structure simultaneously. (4) A fixed vertex of res on $\square^d$ is an N -fixed vertex of $\square^n$ inside the locus — excluded by mixedness. Reflection transport: for a reflection $\tilde{\rho}$ of N , the conjugate $\rho := c \tilde{\rho} c^{-1}$ is a reflection with a fixed vertex v ; the vertex map of F sends v into $\text{Fix}(S)$ with $F(v) = F(\rho v) = \sigma_{\tilde{\rho}}(F(v))$, so $\text{res}(\tilde{\rho})$ fixes the $\square^d$ -coordinates of $F(v)$. Vertex-fixing signed permutations are reflections, and N is generated by its reflections, so res is generated by reflections. $\square$

6. THE MAIN THEOREM

We use from [1], at $\mathcal{V}$ -generality: the trichotomy for the witness machine J_N/N (constant / through the stabilized sieve / strictly invariant) and the stage-zero visibility lemma — their sector and first-exit proofs use only the coequalizer presentation of the algebraic replacement, connections and diagonals, the contractible free body, and the covering classification, all present

$$\begin{array}{ccccc}
& & \bar{g}\bar{j}=[\bar{F}\iota] \text{ twisted } \simeq \text{id} & & \\
& & \downarrow & & \\
\Box^n/N & \xrightarrow{\bar{g}=[\bar{F}]} & \Box^d/\text{res} & \xrightarrow{\bar{j}=[\iota]} & \Box^n/N \\
& \searrow & & \nearrow & \\
& & \text{id} \simeq \bar{j}\bar{g}=[F] & &
\end{array}$$

FIGURE 2. The algebraic reduction (Theorem 5.1): both composites are twisted cells, hence χ -one-step homotopic to identities in both structures — no power trick, no invertibility.

on every reversal site (indeed the machinery of [2] was built on the De Morgan site); the test-side weak equivalence of $\bar{\Phi}_N$; covering descent along free quotients; and the agreement theorem.

Proof of Theorem 1.1. The agreement half and the descent along the free H/N -quotient are as in [1], on every site. For the separation half let N be mixed and reflection-generated; induct on n . If some twisted $\mathcal{V}$ -cell is stabilized, Theorem 5.1 produces an equivalence, in both structures, with $\Box^d/\text{res}$, $d < n$, res mixed reflection-generated; the inductive witness transports across the equivalence by two-out-of-three. If no twisted $\mathcal{V}$ -cell is stabilized, then by Corollary 3.3 the strict component of the identity consists of twisted classes and excludes constants, sieve-landing classes, and strictly invariant classes; by stage-zero visibility this computes the type homotopy relation, and by the trichotomy every lift candidate along $\bar{\Phi}_N$ would put the identity in one of the three excluded classes. Hence the identity does not lift and $\bar{\Phi}_N \in \mathbf{W}_{\text{test}} \setminus \mathbf{W}_{\text{type}}$. (The recursion terminates: d strictly decreases, and no mixed reflection-generated subgroup exists at $n \leq 1$.) $\square$

Remark 6.1. Combined with Proposition 4.2, the *shape* of the recursion also transfers: a class that is rigid on the Boolean site is rigid on $\mathcal{V}$. A Boolean reduction class may a priori be rigid on $\mathcal{V}$ (if its witness system is not $\mathcal{V}$ -realizable) — the theorem is indifferent, since both branches yield separation — but at $n = 4$ this does not happen: see the next section.

7. MACHINE VERIFICATION AND THE $n = 4$ DICTIONARY

Two computations accompany the paper, in the artifact repository of the series (release v1.20.0 and the release accompanying this paper).

The interpolant (`selfdual_interpolant.py`). The three identities of Lemma 3.1 are verified on the literal-cube representation of $F_{\text{DM}}(3)$ (all 64 points), together with monotonicity.

The $n = 4$ reduction classes (`dm_witness_check.py`). For the two Boolean reduction classes of [1] (both of order 8, tree arrangements), existence of a stabilized twisted $\mathcal{V}$ -cell is decided *exactly*: every Boolean witness has vertex

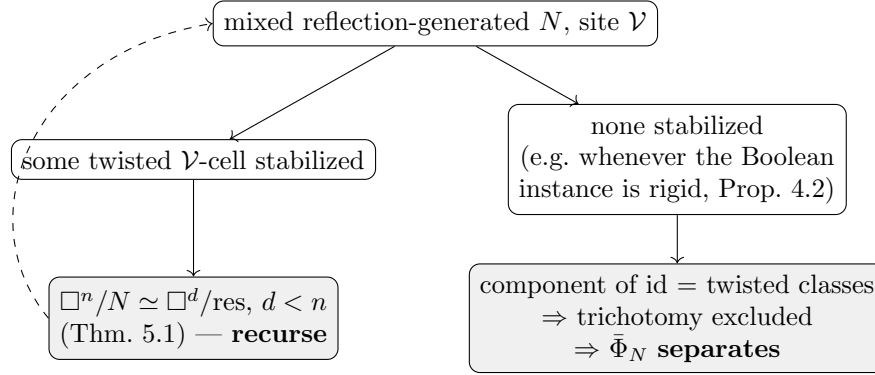


FIGURE 3. The recursion of Theorem 1.1 on an arbitrary reversal site.

image a single invariant plane with pattern factorization, so a $\mathcal{V}$ -witness is a pair (f, g) of free-algebra elements with an exact equivariance system, i.e. a system of parity equalities ($\neg$ acts by $(\neg f)(p) = 1 - f(\rho p)$) and monotonicity implications over the literal cube L_4 (De Morgan; 256 points) or its unmixed subposet U_4 (Kleene) — a 2-SAT instance. The results:

class	invariant plane	DM	KL
first (order 8)	$\{5, 6, 9, 10\}$	witness	witness
second (order 8)	$\{0, 1, 2, 3, 12, 13, 14, 15\}$	—	—
	$\{4, 5, 6, 7, 8, 9, 10, 11\}$	—	—
	$\{0, 3, 4, 7, 8, 11, 12, 15\}$	witness	witness

Both reduction classes admit De Morgan and Kleene witnesses — the second on exactly one of its three invariant planes, the same plane that carries its Boolean witnesses. Hence the $n \leq 4$ classification of [1] transfers branch by branch: 28 rigid classes by Proposition 4.2, 2 reduction classes by Theorem 5.1 with the exhibited witnesses, landing on the Klein quotient $\square^2/K$ and on an order-4 mixed block in dimension three.

Data availability. All verification scripts are archived with the artifact of the series; the release accompanying this paper is version-pinned with a Zenodo DOI recorded on the release page.

Use of AI tools. An AI assistant was used in the preparation of this manuscript and its verification scripts; the author takes full responsibility for the content.

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